

BMS5309 Assessment Task-1

We are working throughout in the `~/amoh0254_BMS5309_AT1/` directory.

1. Quality Control (QC) - FastQC and MultiQC:

FastQC and MultiQC are conducted on all the data files to check the quality of the data. These are done by loading and availing the fastqc and multiqc modules:

```
module load fastqc
```

```
module avail fastqc
```

```
module load multiqc
```

```
module avail multiqc
```

This is done by creating the given shell file using nano:

```
nano run_fastqc.sh
```

and executing the code:

```
#!/bin/bash
input_dir=~ /AT1_data
output_dir=~ /amoh0254_BMS5309_AT1
# Create output directory if it doesn't already exist (-p
#allows creation of nested directories)
mkdir -p "$output_dir"
for file in "$input_dir"/*.fastq
do
echo "$file"
fastqc "$file" --outdir="$output_dir"
done
multiqc "$output_dir" -o "$output_dir"
```

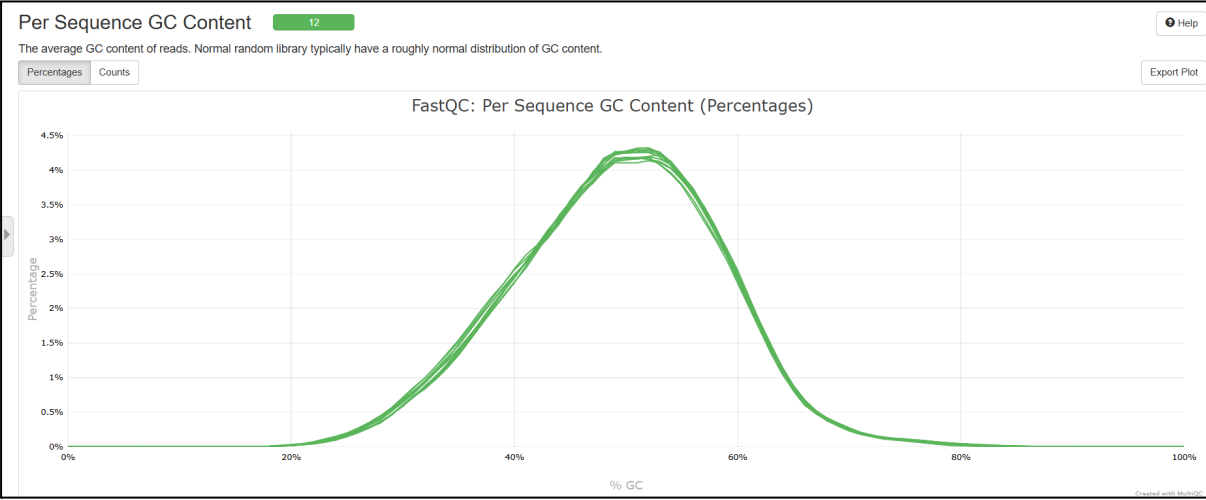
The code is run using:

```
bash run_fastqc.sh
```

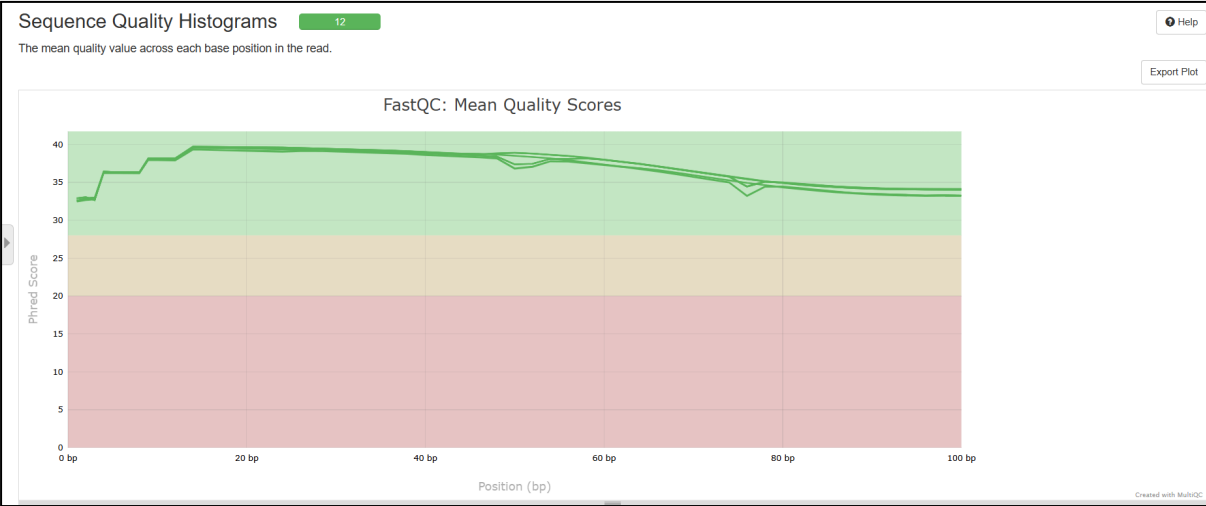
Outputs and Discussion:

General Statistics			
Copy table	Configure columns	Scatter plot	Violin plot
Showing 12/12 rows and 3/4 columns			
Sample Name	% Dups	% GC	M Seqs
naive_stimulated_rep1_R1	9.1%	48%	0.4 M
naive_stimulated_rep1_R2	9.2%	48%	0.4 M
naive_stimulated_rep2_R1	8.5%	48%	0.4 M
naive_stimulated_rep2_R2	8.6%	48%	0.4 M
naive_stimulated_rep3_R1	8.8%	48%	0.4 M
naive_stimulated_rep3_R2	6.5%	48%	0.4 M
naive_unstimulated_rep1_R1	9.1%	48%	0.4 M
naive_unstimulated_rep1_R2	9.2%	48%	0.4 M
naive_unstimulated_rep2_R1	8.1%	48%	0.4 M
naive_unstimulated_rep2_R2	8.3%	48%	0.4 M
naive_unstimulated_rep3_R1	6.8%	48%	0.4 M
naive_unstimulated_rep3_R2	6.2%	48%	0.4 M

The general statistics from the initial MultiQC report show very low proportions of duplicants (%dups), and an ideal %GC of 48% suitable for PCR priming (40% - 60%). The detailed GC content plot is shown below, which also shows an ideal GC content around 40 to 60%:



Moreover, the mean quality score is also shown as ideal in the histogram:



Overall, the sequences seem to be of good quality. Further, we trim the sequences to get better quality score.

2. Adaptor Trimming - Trimmomatic:

This is done to improve the quality of the sequences. This removes low-quality reads, adaptor sequences and poor-quality bases. The sequences are trimmed to improve their quality using the Trimmomatic module:

```
module load trimmomatic

module avail trimmomatic
```

Create an output directory for saving the output files:

```
mkdir trimmomatic
```

This is done by creating the given shell file using nano:

```
nano run_trimmomatic.sh
```

and executing the code:

```
#!/bin/bash
input_dir=~ /AT1_data
output_dir=~ /amoh0254_BMS5309_AT1/trimmomatic
mkdir -p "$output_dir"
for file1 in "${input_dir}"/*_R1.fastq
do
file2="${file1/_R1.fastq/_R2.fastq}"
if [[ ! -f "$file2" ]]; then
echo "Warning: Pair file not found for $file1" continue
fi
# Extract sample base name (e.g., SRR975551)
name=$(basename "$file1" "_R1.fastq")
# Define output file names
out1="${output_dir}/${name}_R1.trim_pe.fastq"
out2="${output_dir}/${name}_R2.trim_pe.fastq"
out1se="${output_dir}/${name}_R1.trim_se.fastq"
out2se="${output_dir}/${name}_R2.trim_se.fastq"
echo "Running Trimmomatic for $name..."
java -jar "$TRIMMOMATIC_PATH"/trimmomatic-0.38.jar PE -threads
4 "$file1" "$file2" "$out1" "$out1se" "$out2" "$out2se"
LEADING:3 TRAILING:3 SLIDINGWINDOW:4:15 MINLEN:50
```

done

The code is run using:

```
bash run_trimmomatic.sh
```

Now, the FastQC and MultiQC commands are run again on the trimmed fastq files, and the new FastQC and MultiQC reports are analysed again to find the improvements.

```
mkdir trimmed_fastqc_original
```

```
nano run_trimmed_fastqc.sh
```

```
#!/bin/bash
input_dir=~/.amoh0254_BMS5309_AT1/trimmomatic
output_dir=~/.amoh0254_BMS5309_AT1/trimmed_fastqc_original
# Create output directory if it doesn't already exist (-p
allows creation of nested directories)
mkdir -p "$output_dir"
for file in "$input_dir"/*.trim_pe.fastq
do
echo "$file"
fastqc "$file" --outdir="$output_dir"
done
multiqc "$output_dir" -o "$output_dir"
```

```
bash run_trimmed_fastqc.sh
```

Outputs and Discussion:

Output after running Trimmomatic:

```
TrimmomaticPE: Completed successfully
Running Trimmomatic for naive_stimulated_rep3...
TrimmomaticPE: Started with arguments:
                                -threads
/home/amoh0254/AT1_data/naive_stimulated_rep3_R1.fastq
/home/amoh0254/AT1_data/naive_stimulated_rep3_R2.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_stimulat
ed_rep3_R1.trim_pe.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_stimulat
ed_rep3_R1.trim_se.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_stimulat
```

```
ed_rep3_R2.trim_pe.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_stimulat
ed_rep3_R2.trim_se.fastq          LEADING:3          TRAILING:3
SLIDINGWINDOW:4:15 MINLEN:50
Quality encoding detected as phred33
Input  Read  Pairs:  401186  Both  Surviving:  394073  (98.23%)
Forward Only Surviving:  5043  (1.26%) Reverse Only Surviving:
1628 (0.41%) Dropped:  442  (0.11%)
TrimmomaticPE: Completed successfully
Running Trimmomatic for naive_unstimulated_rep1...
TrimmomaticPE: Started with arguments:
```

-threads 4

```
/home/amoh0254/AT1_data/naive_unstimulated_rep1_R1.fastq
/home/amoh0254/AT1_data/naive_unstimulated_rep1_R2.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep1_R1.trim_pe.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep1_R1.trim_se.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep1_R2.trim_pe.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep1_R2.trim_se.fastq          LEADING:3          TRAILING:3
SLIDINGWINDOW:4:15 MINLEN:50
Quality encoding detected as phred33
Input  Read  Pairs:  413464  Both  Surviving:  405964  (98.19%)
Forward Only Surviving:  4715  (1.14%) Reverse Only Surviving:
1954 (0.47%) Dropped:  831  (0.20%)
TrimmomaticPE: Completed successfully
Running Trimmomatic for naive_unstimulated_rep2...
TrimmomaticPE: Started with arguments:
```

-threads 4

```
/home/amoh0254/AT1_data/naive_unstimulated_rep2_R1.fastq
/home/amoh0254/AT1_data/naive_unstimulated_rep2_R2.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep2_R1.trim_pe.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep2_R1.trim_se.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep2_R2.trim_pe.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep2_R2.trim_se.fastq          LEADING:3          TRAILING:3
SLIDINGWINDOW:4:15 MINLEN:50
Quality encoding detected as phred33
```

```

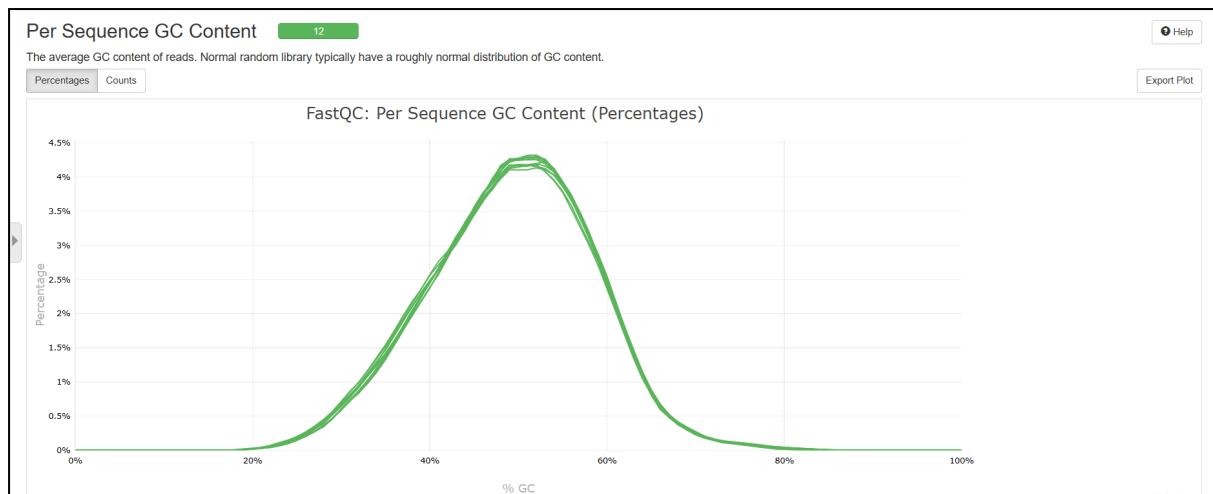
Input Read Pairs: 424313 Both Surviving: 416878 (98.25%)
Forward Only Surviving: 4743 (1.12%) Reverse Only Surviving:
1825 (0.43%) Dropped: 867 (0.20%)
TrimmomaticPE: Completed successfully
Running Trimmomatic for naive_unstimulated_rep3...
TrimmomaticPE: Started with arguments:
                                -threads 4
/home/amoh0254/AT1_data/naive_unstimulated_rep3_R1.fastq
/home/amoh0254/AT1_data/naive_unstimulated_rep3_R2.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep3_R1.trim_pe.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep3_R1.trim_se.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep3_R2.trim_pe.fastq
/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic/naive_unstimul
ated_rep3_R2.trim_se.fastq LEADING:3 TRAILING:3
SLIDINGWINDOW:4:15 MINLEN:50
Quality encoding detected as phred33
Input Read Pairs: 418501 Both Surviving: 410576 (98.11%)
Forward Only Surviving: 6022 (1.44%) Reverse Only Surviving:
1516 (0.36%) Dropped: 387 (0.09%)
TrimmomaticPE: Completed successfully

```

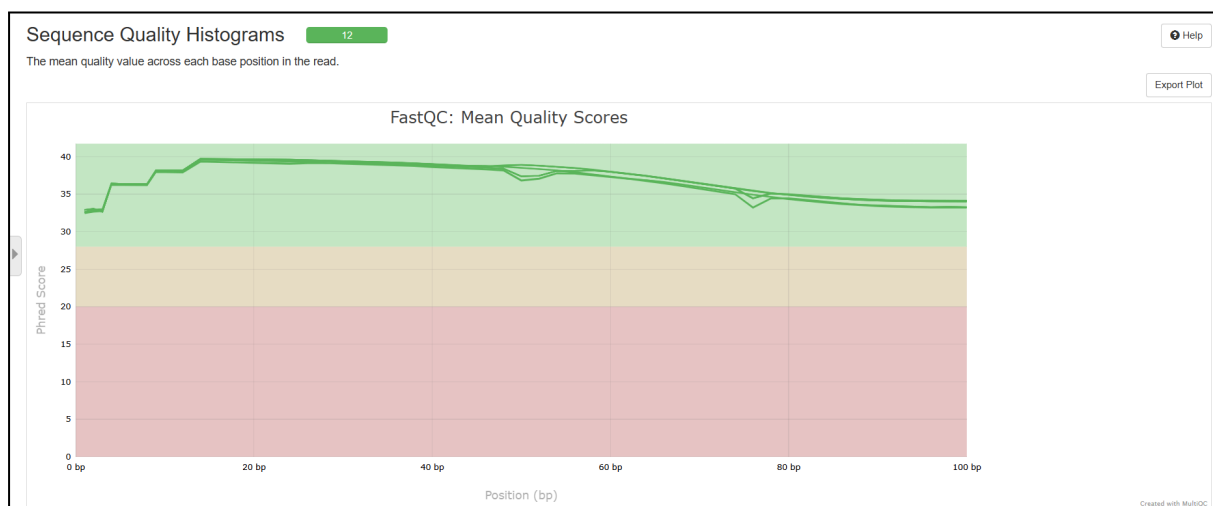
FastQC and MultiQC results:

General Statistics			
Copy table Configure columns Scatter plot Violin plot Showing 12 rows and 3 columns Export as CSV			
Sample Name	% Dups	% GC	M Seqs
naive_stimulated_rep1_R1	9.1%	48%	0.4 M
naive_stimulated_rep1_R2	9.2%	48%	0.4 M
naive_stimulated_rep2_R1	8.5%	48%	0.4 M
naive_stimulated_rep2_R2	8.6%	48%	0.4 M
naive_stimulated_rep3_R1	6.8%	48%	0.4 M
naive_stimulated_rep3_R2	6.5%	48%	0.4 M
naive_unstimulated_rep1_R1	9.1%	48%	0.4 M
naive_unstimulated_rep1_R2	9.2%	48%	0.4 M
naive_unstimulated_rep2_R1	8.1%	48%	0.4 M
naive_unstimulated_rep2_R2	8.3%	48%	0.4 M
naive_unstimulated_rep3_R1	6.8%	48%	0.4 M
naive_unstimulated_rep3_R2	6.2%	48%	0.4 M

There is no significant difference in the general statistics of MultiQC even after trimming. Similar is the case with the GC content graph too:



Similarly, the mean sequence quality score histogram doesn't show much difference:



3. Read Alignment to Reference Genome - HISAT2:

The reads are aligned to the reference genome, which is the mouse reference genome. HISAT2 is used for aligning the reference genome. First, load and avail HISAT2:

```
module load hisat2
```

```
module avail hisat2
```

Create an output directory for saving the output files:

```
mkdir hisat2
```

This is done by again creating the given shell file using nano:

```
nano run_hisat2.sh
```

and again executing the code:

```
#!/bin/bash
reference_genome=/home/amoh0254/mouse_reference/hisat2_ref/mou
se_grcm39_index
input_dir=/home/amoh0254/amoh0254_BMS5309_AT1/trimmomatic
output_dir=/home/amoh0254/amoh0254_BMS5309_AT1/hisat2
mkdir -p "$output_dir"
for file1 in "$input_dir"/*_R1.trim_pe.fastq
do
file2="${file1/_R1.trim_pe.fastq/_R2.trim_pe.fastq}"
if [[ ! -f "$file2" ]]; then
echo "Paired file not found for: $file1"
continue
fi
sample_name=$(basename "$file1"_R1.trim_pe.fastq)
output_sam="${output_dir}/${sample_name}.hisat2.sam"
echo "Aligning sample: $sample_name"
hisat2 -x "$reference_genome" -1 "$file1" -2 "$file2" -S
"$output_sam"
done
```

The code is again run using:

```
bash run_hisat2.sh
```

Outputs and Discussion:

```
2018 mates make up the pairs; of these:
    1617 (80.13%) aligned 0 times
    289 (14.32%) aligned exactly 1 time
    112 (5.55%) aligned >1 times
99.80% overall alignment rate
Aligning sample:
naive_unstimulated_rep2_R1.trim_pe.fastq_R1.trim_pe.fastq
416878 reads; of these:
    416878 (100.00%) were paired; of these:
    877 (0.21%) aligned concordantly 0 times
    394194 (94.56%) aligned concordantly exactly 1 time
    21807 (5.23%) aligned concordantly >1 times
----
```

```

877 pairs aligned concordantly 0 times; of these:
  56 (6.39%) aligned discordantly 1 time
----
821 pairs aligned 0 times concordantly or discordantly; of
these:
  1642 mates make up the pairs; of these:
    1236 (75.27%) aligned 0 times
    314 (19.12%) aligned exactly 1 time
    92 (5.60%) aligned >1 times
99.85% overall alignment rate
Aligning                                                    sample:
naive_unstimulated_rep3_R1.trim_pe.fastq_R1.trim_pe.fastq
410576 reads; of these:
  410576 (100.00%) were paired; of these:
    1571 (0.38%) aligned concordantly 0 times
    388672 (94.67%) aligned concordantly exactly 1 time
    20333 (4.95%) aligned concordantly >1 times
----
  1571 pairs aligned concordantly 0 times; of these:
    122 (7.77%) aligned discordantly 1 time
----
  1449 pairs aligned 0 times concordantly or discordantly;
of these:
    2898 mates make up the pairs; of these:
      1702 (58.73%) aligned 0 times
      895 (30.88%) aligned exactly 1 time
      301 (10.39%) aligned >1 times
99.79% overall alignment rate

```

The overall alignment scores are all above 99%, which is ideal. Originally starting with 12 sequences, now we have 6 aligned pairs of sequences.

4. Conversion of SAM files to BAM files - Samtools

The results from the alignment using HISAT2 are in SAM file format, which is a human-readable format, but BAM file format is computer-readable and more efficient for processing the data. We use the Samtools module for making this conversion, then sorting, indexing and finally getting the summary statistics.

First, load and avail Samtools:

```
module load samtools
```

```
module avail samtools
```

To convert files from SAM to BAM, use:

```
samtools view -b X.sam > X.bam
```

on every SAM file in the hisat2 directory.

Now, the created BAM files are sorted so that the genome is in the right order of occurrence. Each BAM file is sorted using:

```
samtools sort X.bam > X.sorted.bam
```

Now, these sorted BAM files are indexed so that the genes can be marked. This is another essential step, which is done using:

```
samtools index X.sorted.bam
```

Finally, the flagstat command is used to get the summary statistics of each of the sorted and indexed BAM files:

```
samtools flagstat <in.bam>
```

Outputs and Discussion:

```
[amoh0254@m3k015 hisat2]$ samtools flagstat
naive_stimulated_rep1_R1.trim_pe.fastq_R1.trim_pe.fastq.hisat2
.sorted.bam
890796 + 0 in total (QC-passed reads + QC-failed reads)
788226 + 0 primary
102570 + 0 secondary
0 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
889393 + 0 mapped (99.84% : N/A)
786823 + 0 primary mapped (99.82% : N/A)
788226 + 0 paired in sequencing
394113 + 0 read1
394113 + 0 read2
786348 + 0 properly paired (99.76% : N/A)
786670 + 0 with itself and mate mapped
153 + 0 singletons (0.02% : N/A)
182 + 0 with mate mapped to a different chr
129 + 0 with mate mapped to a different chr (mapQ>=5)
```

```
[amoh0254@m3k015 hisat2]$ samtools flagstat
naive_stimulated_rep2_R1.trim_pe.fastq_R1.trim_pe.fastq.hisat2
.sorted.bam
919767 + 0 in total (QC-passed reads + QC-failed reads)
818914 + 0 primary
100853 + 0 secondary
0 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
918475 + 0 mapped (99.86% : N/A)
817622 + 0 primary mapped (99.84% : N/A)
818914 + 0 paired in sequencing
409457 + 0 read1
409457 + 0 read2
817190 + 0 properly paired (99.79% : N/A)
817460 + 0 with itself and mate mapped
162 + 0 singletons (0.02% : N/A)
184 + 0 with mate mapped to a different chr
126 + 0 with mate mapped to a different chr (mapQ>=5)
```

```
[amoh0254@m3k015 hisat2]$ samtools flagstat
naive_stimulated_rep3_R1.trim_pe.fastq_R1.trim_pe.fastq.hisat2
.sorted.bam
874462 + 0 in total (QC-passed reads + QC-failed reads)
788146 + 0 primary
86316 + 0 secondary
0 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
870112 + 0 mapped (99.50% : N/A)
783796 + 0 primary mapped (99.45% : N/A)
788146 + 0 paired in sequencing
394073 + 0 read1
394073 + 0 read2
779612 + 0 properly paired (98.92% : N/A)
780708 + 0 with itself and mate mapped
3088 + 0 singletons (0.39% : N/A)
848 + 0 with mate mapped to a different chr
610 + 0 with mate mapped to a different chr (mapQ>=5)
```

```
[amoh0254@m3k015 hisat2]$ samtools flagstat
naive_unstimulated_rep1_R1.trim_pe.fastq_R1.trim_pe.fastq.hisat2.sorted.bam
919334 + 0 in total (QC-passed reads + QC-failed reads)
```

```
811928 + 0 primary
107406 + 0 secondary
0 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
917717 + 0 mapped (99.82% : N/A)
810311 + 0 primary mapped (99.80% : N/A)
811928 + 0 paired in sequencing
405964 + 0 read1
405964 + 0 read2
809762 + 0 properly paired (99.73% : N/A)
810140 + 0 with itself and mate mapped
171 + 0 singletons (0.02% : N/A)
206 + 0 with mate mapped to a different chr
143 + 0 with mate mapped to a different chr (mapQ>=5)
```

```
[amoh0254@m3k015 hisat2]$ samtools flagstat
naive_unstimulated_rep2_R1.trim_pe.fastq_R1.trim_pe.fastq.hisa
t2.sorted.bam
```

```
922382 + 0 in total (QC-passed reads + QC-failed reads)
833756 + 0 primary
88626 + 0 secondary
0 + 0 supplementary
0 + 0 duplicates
0 + 0 primary duplicates
921146 + 0 mapped (99.87% : N/A)
832520 + 0 primary mapped (99.85% : N/A)
833756 + 0 paired in sequencing
416878 + 0 read1
416878 + 0 read2
832002 + 0 properly paired (99.79% : N/A)
832338 + 0 with itself and mate mapped
182 + 0 singletons (0.02% : N/A)
220 + 0 with mate mapped to a different chr
165 + 0 with mate mapped to a different chr (mapQ>=5)
```

```
[amoh0254@m3k015 hisat2]$ samtools flagstat
naive_unstimulated_rep3_R1.trim_pe.fastq_R1.trim_pe.fastq.hisa
t2.sorted.sam
```

```
898777 + 0 in total (QC-passed reads + QC-failed reads)
821152 + 0 primary
77625 + 0 secondary
0 + 0 supplementary
0 + 0 duplicates
```

```
0 + 0 primary duplicates
897075 + 0 mapped (99.81% : N/A)
819450 + 0 primary mapped (99.79% : N/A)
821152 + 0 paired in sequencing
410576 + 0 read1
410576 + 0 read2
818010 + 0 properly paired (99.62% : N/A)
819020 + 0 with itself and mate mapped
430 + 0 singletons (0.05% : N/A)
834 + 0 with mate mapped to a different chr
617 + 0 with mate mapped to a different chr (mapQ>=5)
```

The flagstat reports show that all reads passed quality control, above 99% of the reads are properly paired, and there are no primary duplicates. All these are signs of good alignment and quality control.

5. Mapping and Quantification of Transcripts with FeatureCounts - featureCounts:

The featureCounts module is used to assess and count the features of the genome. The genome has various features like SNPs, exons, introns, etc. and all these can be identified, counted and summarised by featureCounts. First, load and avail subread for featureCounts:

```
module load subread
```

```
module avail subread
```

Create an output directory for saving the output files:

```
mkdir feature_counts
```

This is done by again creating the given shell file using nano:

```
nano run_feature_counts.sh
```

and again executing the code:

```
#!/bin/bash
```

```
reference_gtf="/home/amoh0254/mouse_reference/Mus_musculus.GRC
m39.110.gtf"
bam_dir="/home/amoh0254/amoh0254_BMS5309_AT1/hisat2"
```

The figure shows a snapshot of the featureCounts result, which specifies the different exons in the sequences. The summary file shows a summarised view of the featureCounts results:

Status	/home/amoh0254/amoh0254_BMS5309_AT1/hisat2/naive_stimulated_rep1_R1.trim_pe.fastq_R1.trim_pe.fastq.hisat2.sorted.bam	/home/amoh0254/amoh0254_BMS5309_AT1/hisat2/naive_stimulated_rep2_R1.trim_pe.fastq_R1.trim_pe.fastq.hisat2.sorted.bam	/home/amoh0254/amoh0254_BMS5309_AT1/hisat2/naive_stimulated_rep3_R1.trim_pe.fastq_R1.trim_pe.fastq.hisat2.sorted.bam	/home/amoh0254/amoh0254_BMS5309_AT1/hisat2/naive_unstimulated_rep1_R1.trim_pe.fastq_R1.trim_pe.fastq.hisat2.sorted.bam	/home/amoh0254/amoh0254_BMS5309_AT1/hisat2/naive_unstimulated_rep2_R1.trim_pe.fastq_R1.trim_pe.fastq.hisat2.sorted.bam
Assigned	623672	650905	629968	610076	626036
Unassigned_Unmapped	1403	1292	4350	1617	1236
Unassigned_Read_Type	0	0	0	0	0
Unassigned_Singleton	0	0	0	0	0
Unassigned_MappingQuality	0	0	0	0	0
Unassigned_Chimera	0	0	0	0	0
Unassigned_FragmentLength	0	0	0	0	0
Unassigned_Duplicate	0	0	0	0	0
Unassigned_MultiMapping	149422	147379	130338	155206	132332
Unassigned_Secondary	0	0	0	0	0
Unassigned_NonSplit	0	0	0	0	0
Unassigned_NoFeatures	84975	88282	82410	120821	131349
Unassigned_Overlapping_Length	0	0	0	0	0
Unassigned_Ambiguity	31324	32309	31396	30814	31429

6. Summary and Critical Evaluation:

In the given RNA-seq project, the fastq files of the sequences were first analysed for their quality using FastQC and MultiQC, which gave good quality scores. They were trimmed for better quality scores. Later, they were aligned using HISAT2, and then indexed. This gave us the aligned sequences for further analysis of feature counts to understand the exon regions of the sequences.

7. AI Acknowledgement:

I acknowledge the use of Google Gemini (<https://gemini.google.com/app?hl=en-AU>) for understanding basic concepts related to the assessment.